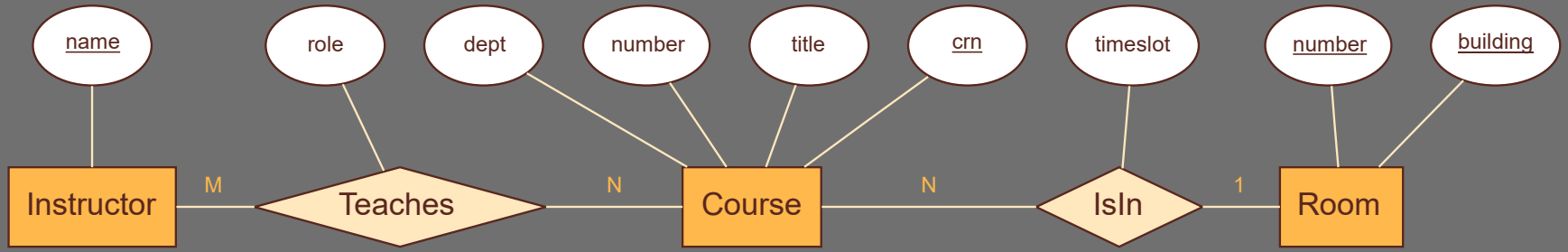
ERD Building Blocks

Entity sets

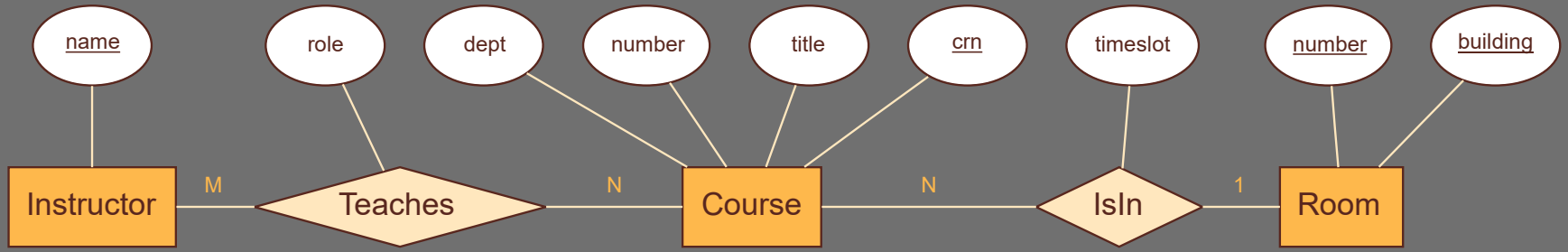


- Entity sets
- Relationships
- Attributes

Instructor

- Finite sets of individual entities/items
- Each entity is mapping from attribute names to values
- Each key value is unique

Relationships

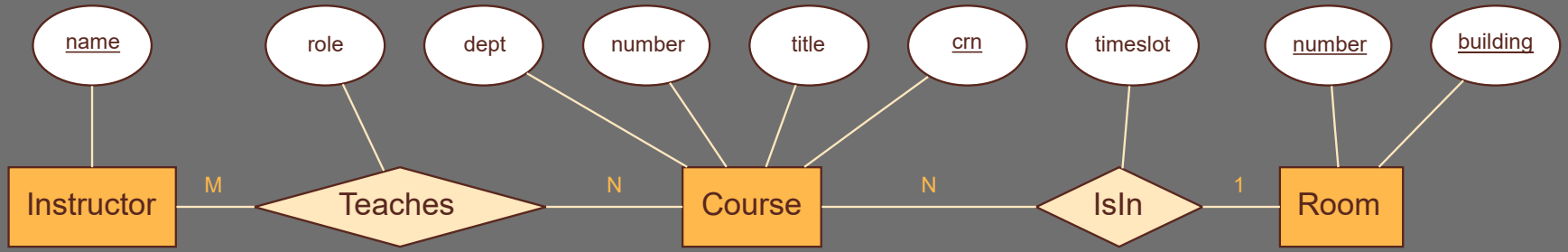


- Entity sets
- Relationships
- Attributes



- Subset of cross product of entity sets
- Key is cross product of entity sets' keys.

Attributes



- Entity sets
- Relationships
- Attributes



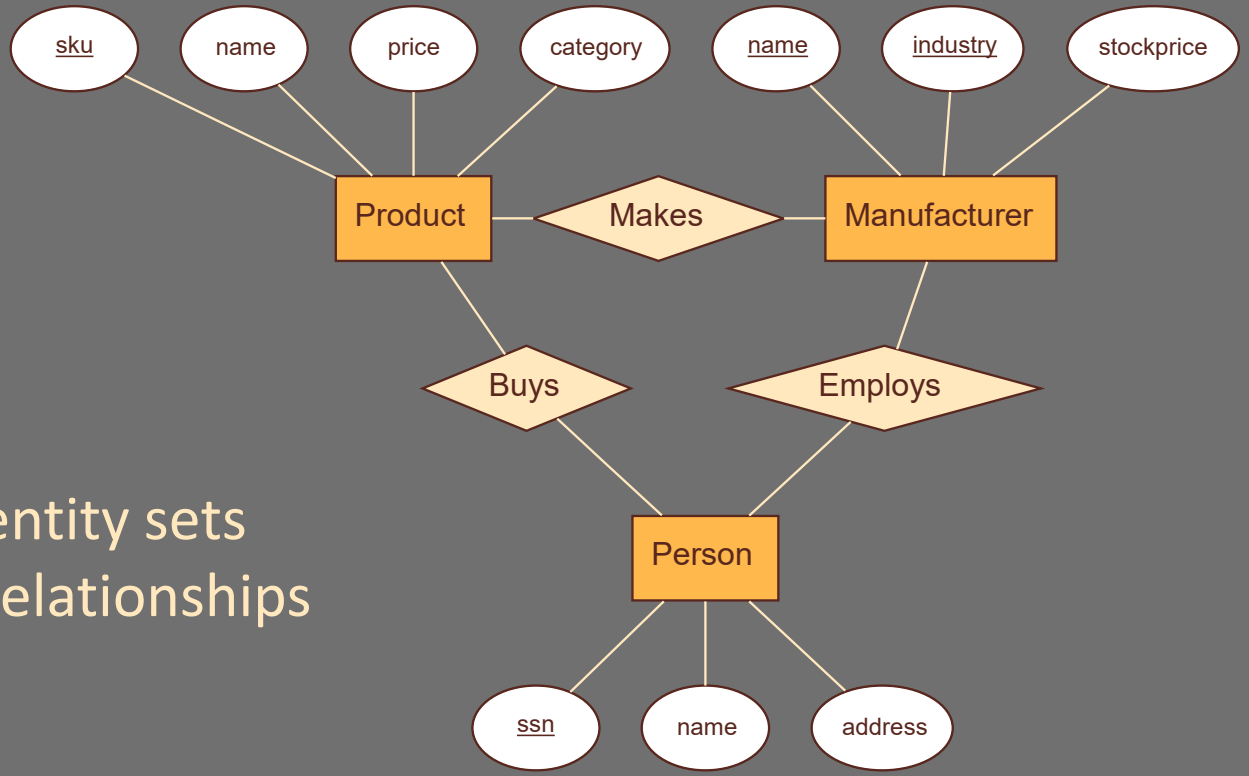
- Elaborate entity sets or relationships

ERDs ≠ SQL

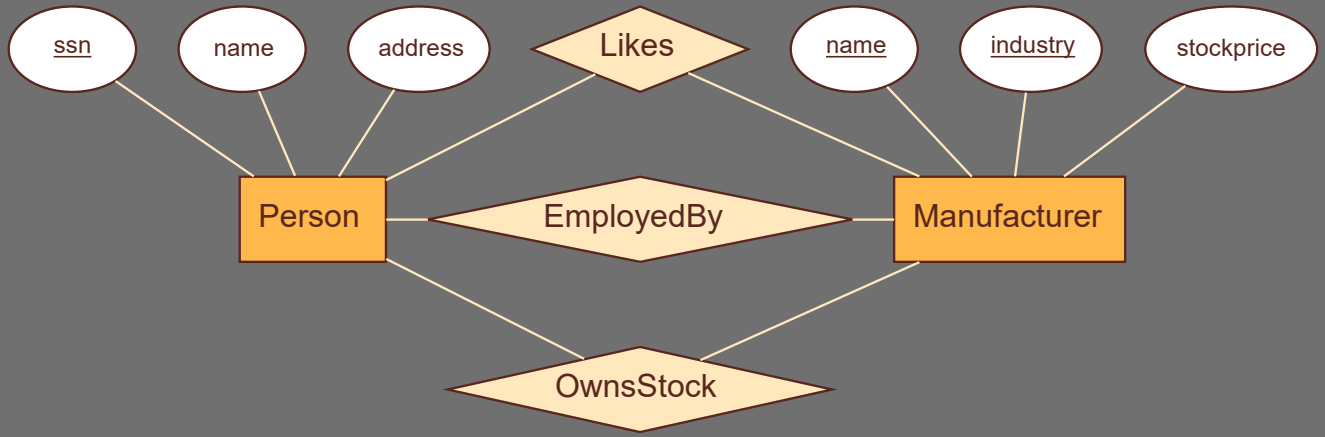
- Entity sets ≠ Tables
- Relationships ≠ Tables

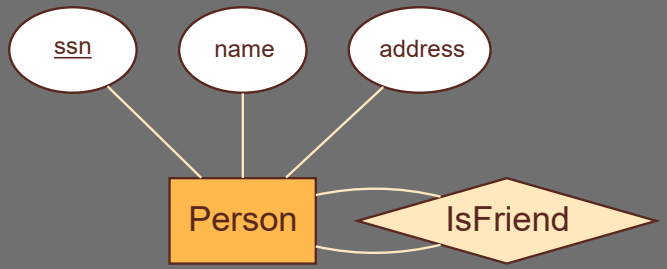
- Attributes include (primary) keys.
- Attributes do not include FKs.

SAMPLE COMBINATIONS



Many entity sets
Many relationships


ssn
name
address
Likes
name
industry
stockprice
Person
EmployedBy
Manufacturer
OwnsStock


ssn
name
address
Person
IsFriend

N-ary relationships

